EMTH210

LECTURE 40:

Announcement:

Recaps last lecture

7.3.1 Marginal Distributions

Given a bivariate random vector (X, Y) with PDF $f_{X,Y}$, the marginal distribution of X describes the probability density of X irrespective of the value Y. We say "Y has been marginalised out".

where both X/Y are 0 and 1

x/y	0	1
0	0	0.2
1	0.3	0.5

Marginal PDF: The marginal PDF for X is

$$f_X(x) = \int_{-\infty}^{+\infty} \underbrace{f_{X,Y}(\overset{\text{fixed}}{x}, y)}_{\text{function of } y}\,\mathrm{d}y,$$

and the marginal PDF for Y is

$$f_Y(y) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y)\,\mathrm{d}x.$$

For example:

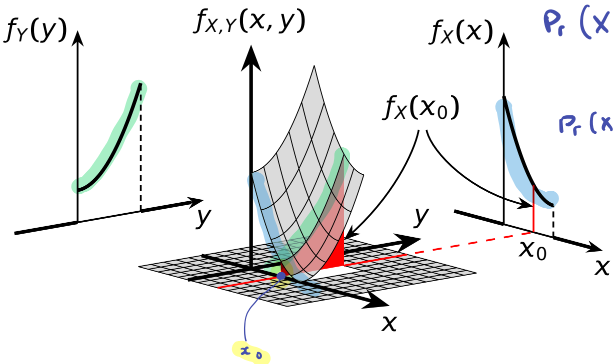
Probability → $P_r((X,Y) = (0,1)) = 0.2$

we dont care what Y is

$$P_r(X=1) = P_r\big(\underbrace{(X,Y) = (1,0) \text{ or } (X,Y) = (1,1)}\big) = 0.3 + 0.5 = 0.8$$

$P_r(X=0) = 0.2$

For now think what probability we are going to get x.

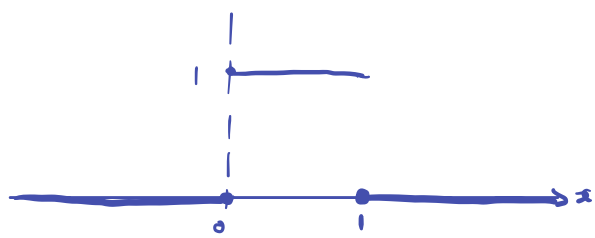


Exercise 7 Marginals

Find the marginal PDFs $f_X(x)$ and $f_Y(y)$ from the PDF in Exercise 6 (p29).

$$f_X(x) = \int_{-\infty}^{\infty} f_{x,y}(x,y)\,dy = \begin{cases} 0 & \text{if } x < 0 \text{ or } x > 1 \\ \int_0^1 1\,dy = 1 & \text{otherwise} \end{cases}$$

$$f_{x,y}(x,y) = \begin{cases} 1 & \text{if } (x,y) \in [0,1]^2 \\ 0 & \text{otherwise} \end{cases}$$



$f_X(x)$

X is uniform on [0, 1

7.3.2 Independence of Random Variables

In EMTH119 we covered independence of *events*. Here we extend this idea to random variables. Two random variables are said to be independent if and only if

$$f_{X,Y}(x,y) = f_X(x) f_Y(y) \longleftarrow$$

for every $(x,y) \in \mathbb{R}^2$, or equivalently

$$F_{X,Y}(x,y) = F_X(x) F_Y(y)$$

for every $(x,y) \in \mathbb{R}^2$. Independence can arise in two distinct ways. We either explicitly assume that random variables are independent (think of tossing a coin twice, the coin has no memory of the first toss) or we derive independence.

Intuition

First sample X according to $f_X(x)$ (marginal PDF for X)
Then, completely ignoring X, sample Y according to $f_X(y)$
for the distribution of such (X', Y') is the same as
the distribution of (X, Y), they are independent

Exercise 8 Independence

Let (X,Y) be the uniform random vector introduced in Exercise 6 (p29). Are X and Y independent?

$$f_X(x) = \begin{cases} 1 & \text{if } x \in [0,1] \\ 0 & \text{otherwise} \end{cases} \qquad f_Y(y) = \begin{cases} 1 & \text{if } y \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x) f_Y(y) = \left.\begin{cases} 1 & \text{if } (x,y) \in [0,1]^2 \\ 0 & \text{otherwise} \end{cases}\right\} = f_{X,Y}(x,y)$$

Answer: Yes they are independent

Exercise 9 PDF From Marginals

If X and Y are independent random variables with PDFs defined below, what is their joint probability density function?

$$f_X(x) = \begin{cases} 2e^{-2x} & \text{if } x > 0 \\ 0 & \text{otherwise,} \end{cases} \qquad f_Y(y) = \begin{cases} 3e^{-3y} & \text{if } y > 0 \\ 0 & \text{otherwise.} \end{cases}$$

$$f_{X,Y}(x,y) = f_X(x) \times f_Y(y) = \begin{cases} 6e^{-2x-3y} & \text{if } x,y > 0 \\ 0 & \text{otherwise} \end{cases}$$

7.3.3 Expectation with Multiple Random Variables

The expectation of a function $g(X,Y)$ of a random vector (X,Y) is defined as

$$\mathrm{E}\big(g(X,Y)\big) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} g(x,y) f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y.$$

The expected values of X and Y are found by setting $g(X,Y) = X$ and $g(X,Y) = Y$, respectively. If X and Y are **independent** random variables, then the following properties hold:

1. $\mathrm{E}(XY) = \mathrm{E}(X)\,\mathrm{E}(Y)$
2. $\mathrm{E}\big(h_1(X)\,h_2(Y)\big) = \mathrm{E}\big(h_1(X)\big)\,\mathrm{E}\big(h_2(Y)\big)$
   (h_1 is a function of X, h_2 is a function of Y)
3. $\mathrm{Var}(aX + bY + c) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y)$

If X and Y are **independent or dependent** random variables, then the following properties hold:

1. $\mathrm{E}\big(a\,g(X,Y)\big) = a\,\mathrm{E}\big(g(X,Y)\big)$, for $a \in \mathbb{R}$
2. $\mathrm{E}\big(g(X,Y) + h(X,Y)\big) = \mathrm{E}\big(g(X,Y)\big) + \mathrm{E}\big(h(X,Y)\big)$
3. $\mathrm{E}(aX + bY + c) = a\,\mathrm{E}(X) + b\,\mathrm{E}(Y) + c$, for $a, b, c \in \mathbb{R}$
4. $\mathrm{Var}(aX + bY + c) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X,Y)$, for $a, b \in \mathbb{R}$,
   where $\mathrm{Cov}(X,Y)$ is the covariance of X and Y (defined in the next section).

EXPECTATION

Linearity of expectation $E(X+Y) = E(X) + E(Y)$

$$E(aX) = \int_{-\infty}^{\infty} ax\, f_X(x)\, dx = a\int_{-\infty}^{\infty} x f_X(x)\, dx = a\,E(X)$$

Always!

$a \in \mathbb{R}$

If X and Y are independent, then

$E(XY) = E(X)E(Y)$

PROOF:

$$E(XY) = \int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty} xy\, \underbrace{f_{X,Y}(x,y)}_{= f_X(x) f_Y(y)}\, dy\right) dx = \int_{-\infty}^{\infty}\left(x f_X(x)\, \overbrace{\int_{-\infty}^{\infty} y f_Y(y)\, dy}^{E(Y)}\right) dx$$

$$= E(Y)\underbrace{\int_{-\infty}^{\infty} x f_X(x)\, dx}_{E(X)} = E(Y)E(X)$$

$\boxed{\text{Exercise}}$ Come up with $f_{X,Y}(x,y)$ such that $E(XY) \neq E(X)E(Y)$

VARIANCE:

$$Var(X) = E\left((X - E(X))^2\right) = \underbrace{E(X^2) - E(X)^2}_{\text{easier to sample}}$$

→ What happens when we scale with 'a'?

- $Var(aX) = E(aX)^2 = E(a^2X^2) - a^2E(X)^2 = a^2E(X^2) - a^2E(X)^2 = a^2(E(X^2) - E(X)^2) = a^2 Var(X)$
  $a \in \mathbb{R}$

- $E(X+a) = E(X) + E(a) = E(X) + a$

$Var(X+a) = Var(X)$

- If X and Y are independent, then
  $Var(X+Y) = Var(X) + Var(Y)$ ← NOT TRUE in general